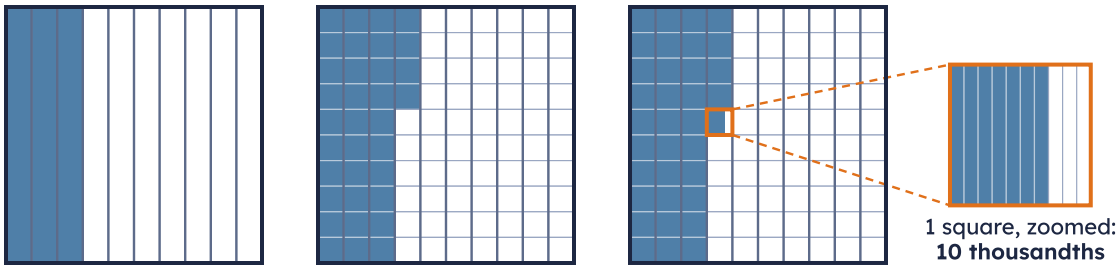
GABRIELLA'S MATH LAB · STUDY GUIDE

DECIMALS · LESSON 1

Understanding Decimals

Tenths, hundredths, and thousandths: what the digits after the decimal point really mean.



0.3

3 **tenths**

0.34

34 **hundredths**

0.347

347 **thousandths**

Inside this guide

- Learn: goals, words, and the big idea 2
- See It: pictures of decimals 4
- Worked Examples, step by step 6
- Try It Together (guided practice) 8
- On Your Own (independent practice) 10
- Answer Key with explanations 13
- Parent Mastery Checklist 15

How to use it

1. **Read** the Learn and See It pages with a grown-up. Say the examples out loud.
2. **Follow** each Worked Example, one step at a time.
3. **Try It Together**, using the hints. Then work **On Your Own** with no help.
4. **Check** with the Answer Key and fix any mistakes.

Name _____ Date started _____

Georgia K-12 Mathematics Standards: 5.NR.4.1 (read and write decimals to the thousandths place; expanded form introduced), 5.NR.4.2 (represent decimals using the meaning of the digits in each place; comparing introduced). Connects to 5.NR.1.1 (each place is 10 times the place to its right). Matches Math Lab lesson D01.

What You Will Learn

By the end of this lesson, you should be able to say each of these sentences and prove it.

- [ ] **I can** explain that a whole can be cut into 10, 100, or 1,000 equal parts called tenths, hundredths, and thousandths.
- [ ] **I can** name the place and the value of each digit in a decimal.
- [ ] **I can** explain why 0.5 = 0.50, and why the zero in 0.05 matters.
- [ ] **I can** write the decimal for a shaded model, and shade a model to show a decimal.
- [ ] **I can** read and write decimals in words.
- [ ] **I can** use place value to compare decimals like 0.4 and 0.35.

Words to Know



whole

One complete thing or amount, like a whole bar or a whole grid. One whole is the number **1**.

2.35

decimal

A number that uses a decimal point and place value to show parts of a whole, like **0.7** or **2.35**.

4.6

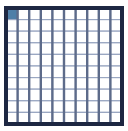
decimal point

The dot that separates the whole number (on the left) from the parts of a whole (on the right). Read it as **"and."**



tenth

One of **10** equal parts of a whole. One tenth is written **0.1** (or $^{1}/_{10}$).



hundredth

One of **100** equal parts of a whole. One hundredth is written **0.01** (or $^{1}/_{100}$).

0.001

thousandth

One of **1,000** equal parts of a whole. Cut one hundredth into 10 equal slices: each slice is **0.001**.

0.627

place value

The value a digit has because of its **place**. In 0.627, the 2 is in the hundredths place, so it is worth **0.02**.

0.5=0.50

equivalent decimals

Decimals that name the **same amount**. 0.5 and 0.50 are equivalent: 5 tenths is the same as 50 hundredths.

Say it with place names: say 0.7 as "seven tenths" instead of "point seven." Saying the place name tells you what the number means.

What Is a Decimal?

Decimals use place value to show wholes and parts of a whole: tenths, hundredths, and thousandths.

START WITH A CHOCOLATE BAR

This chocolate bar is **one whole**. It is made of **10 equal pieces**, so each piece is **one tenth** of the bar.

Gabriella eats **3 pieces**. She ate **3 tenths** of the bar. As a decimal, that is **0.3**. As a fraction, it is $^{3}/_{10}$.

The pieces must be **equal**. If some pieces were bigger, they would not be tenths.



3 of the 10 equal pieces are eaten (dashed). 7 tenths are left.

One whole, cut into smaller and smaller equal parts

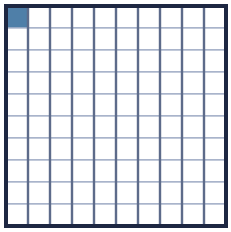
10 EQUAL PARTS



Each part is **one tenth**.

0.1 10 tenths = 1 whole

100 EQUAL PARTS

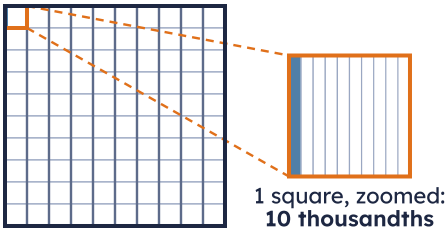


Each small square is **one hundredth**.

0.01

Each column is 1 tenth.

1,000 EQUAL PARTS



Cut each square into 10 slices. Each slice is **one thousandth**.

0.001

The big idea: every place is worth **10 times** the place to its right.

10 thousandths = 1 hundredth → 10 hundredths = 1 tenth → 10 tenths = 1 whole

Why this matters. Look at the hundredths grid: one column has 10 small squares. So **10 hundredths fill exactly 1 tenth**. That is why **0.1 = 0.10**. Going the other way, each place is **one tenth** ($^{1}/_{10}$) of the place to its left.

Where you see decimals: money ($2.35), running distances (0.8 mile), science measurements (0.125 liter), and sports times (9.58 seconds).

Every Digit Has a Place

A place-value chart shows what each digit is worth.

THE NUMBER 0.627 IN A PLACE-VALUE CHART

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
0	.	6	2	7

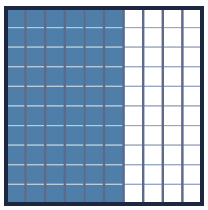
6 is in the **tenths** place.
6 tenths = **0.6**

2 is in the **hundredths** place.
2 hundredths = **0.02**

7 is in the **thousandths** place.
7 thousandths = **0.007**

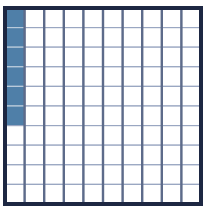
Put the values together: **0.627 = 0.6 + 0.02 + 0.007**. Writing a number as the sum of its place values is called **expanded form**.

Same digit, different place, different value



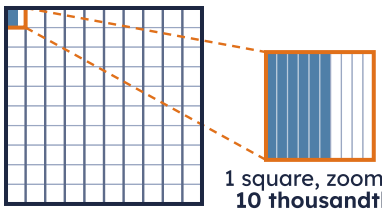
0.6

6 tenths
6 full columns



0.06

6 hundredths
6 small squares



0.006

6 thousandths
6 slices of one square

Each time the 6 moves **one place to the right**, it is worth **one tenth as much**.

HOW TO READ A DECIMAL OUT LOUD

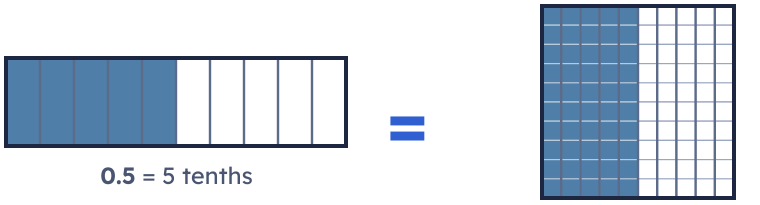
1. If there is a whole number, read it, then say **"and"** for the decimal point.
2. Read the digits after the decimal point like a whole number.
3. Say the place name of the **last** digit.

Decimal	Read it as
0.43	forty-three **hundredths**
0.07	seven **hundredths**
0.627	six hundred twenty-seven **thousandths**
2.5	two **and** five **tenths**

Zeros: When They Matter

Some zeros change a decimal. Some do not. Pictures show which is which.

EQUIVALENT DECIMALS: 0.5 = 0.50



0.5 = 5 tenths

0.50 = 50 hundredths

The same amount is shaded. **5 tenths = 50 hundredths.** A zero added at the **end**, after the decimal digits, does not change the value: 0.5 = 0.50 = 0.500.

A PLACEHOLDER ZERO: 0.05 IS NOT 0.5



0.05 = 5 hundredths

0.5 = 5 tenths

In 0.05, the zero is a **placeholder zero** in the tenths place. It holds the place so the 5 stays in the **hundredths** place. Leave that zero out and the 5 slides into the tenths place: a number **10 times as big**.

WATCH OUT: IS 0.35 BIGGER THAN 0.4?

Some students think **0.35 > 0.4** because 35 is bigger than 4. Look at the grids:



0.4 = 0.40
40 hundredths

0.35
35 hundredths

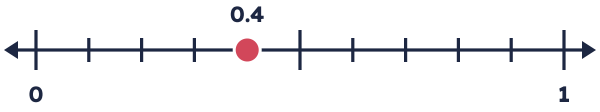
Step 1. Write 0.4 as **0.40** so both numbers have hundredths.

Step 2. Compare: 40 hundredths > 35 hundredths.

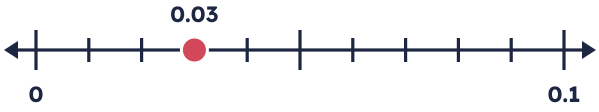
0.4 > 0.35. The tenths agree: 4 tenths > 3 tenths. More digits does **not** mean a bigger decimal.

DECIMALS ON A NUMBER LINE

Find the size of **one jump**, then count the jumps from the start. 1 whole in 10 jumps: **tenths**. 1 tenth in 10 jumps: **hundredths**.



0 to 1: each jump is **0.1**. 4 jumps = **0.4**



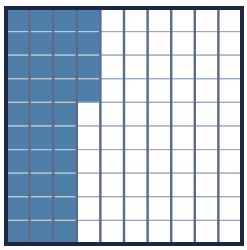
0 to 0.1: each jump is **0.01**. 3 jumps = **0.03**

Step by Step

EXAMPLE 1 · WRITE THE DECIMAL FOR A MODEL

What decimal names the shaded part of the grid?

In every grid in this guide, shading fills one column at a time, top to bottom.

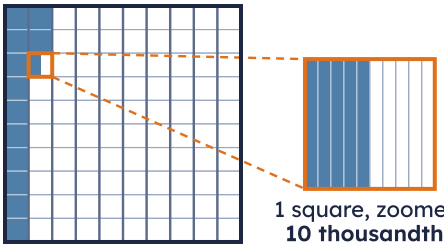


1. The grid is one whole cut into **100** equal squares, so each square is **one hundredth**.
2. Count the **full columns**. Each column is 1 tenth. There are **3** full columns: **3 tenths**.
3. Count the **extra squares**. There are **4**: **4 hundredths**.
4. Write the digits in their places: **0.34**
5. **Check:** 34 of the 100 squares are shaded. Read it as "thirty-four hundredths." ✓ *34 squares!*

EXAMPLE 2 · SHOW THOUSANDTHS ON A GRID

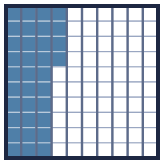
Show 0.125 on a hundredths grid.

1. Find each digit's value: **1 tenth**, **2 hundredths**, **5 thousandths**.
2. Shade **1 full column** for 1 tenth.
3. Shade **2 more squares** for 2 hundredths. That is 12 squares in all.
4. Thousandths are smaller than one square. Cut the **next square** into 10 equal slices and shade **5 slices**.
5. **Check:** 0.125 is a little more than 12 hundredths and less than 13 hundredths. *12 squares + half a square* ✓



Orange box: the square cut into 10 thousandths

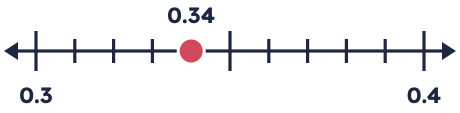
ONE NUMBER, FIVE WAYS: 0.34



grid

Ones 1		Tenths 0.1	Hundredths 0.01
0	.	3	4

place-value chart



number line from 0.3 to 0.4 (each jump is 0.01)

In words: **thirty-four hundredths** · Expanded form: **0.3 + 0.04**

WORKED EXAMPLES Follow each step

EXAMPLE 3 · FIND THE VALUE OF A DIGIT

What is the value of the 8 in 2.385?

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
2	.	3	8	5

1. Write the number in a place-value chart. Line the decimal point up with the chart's point.
2. The 8 is the **second** digit after the decimal point: the **hundredths** place.
3. So the 8 is worth **8 hundredths**, which is **0.08**.

The other digits: 2 ones (2), 3 tenths (0.3), 5 thousandths (0.005). So 2.385 = 2 + 0.3 + 0.08 + 0.005.

EXAMPLE 4 · WRITE A DECIMAL FROM WORDS

Write "seven hundredths" as a decimal.

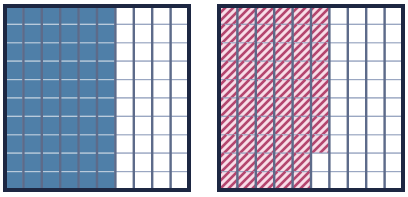
1. The last word is **hundredths**, so the last digit goes in the hundredths place: 2 places after the point.
2. Put the 7 in the hundredths place. The tenths place is empty, so it gets a **placeholder zero**.
3. Answer: **0.07** *not 0.7!*

Ones 1		Tenths 0.1	Hundredths 0.01
0	.	0	7

EXAMPLE 5 · COMPARE TWO DECIMALS

Which is greater, 0.6 or 0.58?

1. Line up the decimal points. Give both numbers the same number of places: **0.6 = 0.60**.
2. Compare from the left. Ones: 0 and 0, the same. Tenths: **6 > 5**.
3. The tenths decide it, so **0.6 > 0.58**. The grids agree: 60 squares > 58 squares.



0.60 0.58

Explain it out loud. Cover the examples, then explain to a grown-up:

- Why is the 8 in 2.385 worth 0.08, not 8?
- Why does "seven hundredths" need a zero right after the decimal point?
- Why is it fair to write 0.6 as 0.60 before comparing it with 0.58?

Try It Together

Work with a grown-up. Fill in the blanks, use a hint only if you need it, and explain each answer using tenths, hundredths, or thousandths.

A

MODELS

What decimal names the shaded part of the bar?



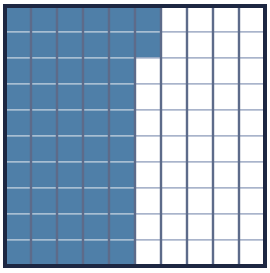
_____ of 10 equal parts are shaded = _____ tenths = _____

Hint Count the shaded parts. Each part is one tenth.

B

MODELS

What decimal names the shaded part of the grid?



_____ full columns = _____ tenths

_____ more squares = _____ hundredths

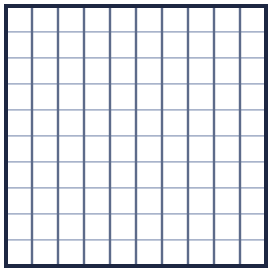
Decimal: _____

Hint Count full columns first. Each column is 10 squares, or 1 tenth.

C

MODELS

Shade the grid to show **0.09**.



0.09 has _____ tenths and _____ hundredths.

So I shade _____ small squares.

Hint Is 0.09 more or less than one full column (0.10)?

D

PLACE VALUE

Write **4.716** in the chart. Then answer.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
	.			

The 7 is in the _____ place.

The value of the 7 is _____

Thousandths digit: _____

Hint The first place after the decimal point is tenths, then hundredths, then thousandths.

E READING AND WRITING DECIMALS

a. Write **0.38** in words. The last digit is in the _____ place.

0.38 is read as _____

b. Write **"two and six tenths"** as a decimal. _____

Hint "And" is the decimal point. The whole number goes on the left.

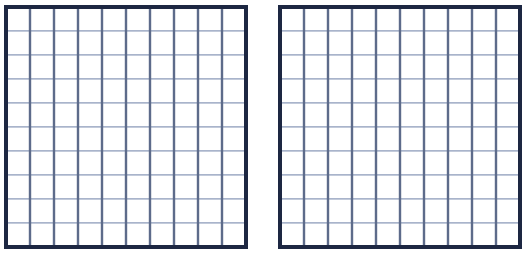
F COMPARING

Which is greater, **0.3** or **0.27**?

Write 0.3 with hundredths: 0.3 = _____

_____ hundredths compared with 27 hundredths

So 0.3 ☐ 0.27 (write >, <, or =)



0.3 0.27

Hint Shade both grids. Which shows more?

G MODELS

The number line from 0 to 1 is cut into 10 equal jumps. What decimal is at point **P**?

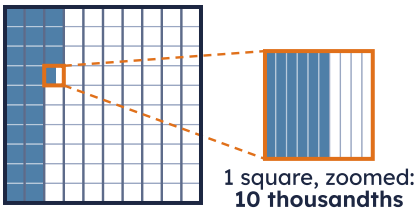


Each jump is _____. P is _____ jumps from 0. P = _____

Hint 10 equal jumps make 1 whole, so each jump is one tenth.

H MODELS WITH THOUSANDTHS

What decimal does this grid show? The orange square is cut into 10 equal slices.



_____ full columns = _____ tenths

_____ more squares = _____ hundredths

_____ slices = _____ thousandths

Decimal: _____

Hint Each slice is one thousandth: 10 slices fill one small square.

On Your Own

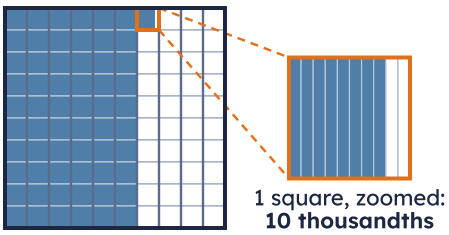
No hints this time. Show your thinking. Then check the Answer Key.

1 A whole is cut into **1,000** equal parts. What is the name of **one** of those parts?

(A) one tenth (B) one hundredth (C) one thousandth (D) one whole

2 What decimal does the grid show? The orange square is cut into 10 equal slices.

Answer: _____



3 Which decimal names the **shaded** part of the bar?



(A) 0.03 (B) 0.3 (C) 3.0 (D) 0.7

4 This number line goes from 0 to **0.1** and is cut into 10 equal jumps. What decimal is at point **A**?

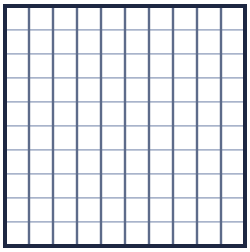


A = _____

5 **Shade the grid** to show **0.45**.

Then write how many full columns and extra squares you shaded.

Full columns: _____ Extra squares: _____



6 In the number **5.438**, which digit is in the **hundredths** place?

(A) 5 (B) 4 (C) 3 (D) 8

MY WORK

ON YOUR OWN Independent practice

7 What is the **value** of the digit 9 in **0.392**? Write it as a decimal.

MY WORK

Answer: _____

8 **a.** Write the decimal that has **6 tenths, 0 hundredths, and 4 thousandths**. Use the chart to help.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
	.			

Decimal: _____

b. Write your decimal in **expanded form**: _____

9 Write **0.209** in words.

10 Write **"three and fifteen hundredths"** as a decimal.

MY WORK

Answer: _____

11 Which decimals are **equal to 0.6**? **Choose all** that are correct.

A 0.06 B 0.60 C 0.600 D 6.0

12 Mia says, *"0.8 and 0.08 are equal, because zeros don't change a decimal."* Is Mia correct? Explain using place value or a model.

MY WORK

13 Write >, <, or = in each box.

a. 0.7 ☐ 0.69

b. 0.25 ☐ 0.250

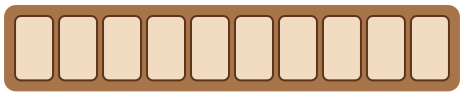
c. 0.04 ☐ 0.4

d. 0.512 ☐ 0.52

14 Jordan ran **0.8 mile**. Ava ran **0.75 mile**. Ava says she ran farther because 75 is greater than 8. **Who ran farther?** Explain using place value or a model.

MY WORK

15 A chocolate bar has 10 equal pieces. Sam eats **4 pieces**. Cross out the 4 pieces Sam eats. What decimal names the part of the bar that is **left**?



Answer: _____

16 Fill in the blanks: 1 tenth = _____ hundredths = _____ thousandths

17 CHALLENGE

What number am I? I am less than 1 and have 3 digits after the decimal point. My thousandths digit is 2. My hundredths digit is 0. My tenths digit is 5 more than my thousandths digit.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
	.			

I am _____

MY WORK

Done? Check every answer in the Answer Key. For each one you missed, read the explanation and fix it in a different color.

Answer Key

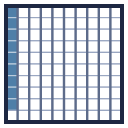
Every answer comes with a reason. Read the reason, even when you got it right.

Try It Together

A **0.7**. 7 of 10 equal parts are shaded: 7 tenths = 0.7.

B **0.52**. 5 full columns = 5 tenths; 2 more squares = 2 hundredths. 52 of 100 squares are shaded.

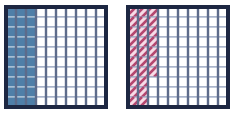
C **Shade 9 small squares** (not a full column). 0.09 = 0 tenths and 9 hundredths, which is less than one column (0.10).



D Chart: 4 . 7 1 6. The 7 is in the **tenths** place, so its value is **0.7** (7 tenths). The thousandths digit is **6.**

E **a.** Last digit is in the hundredths place: **thirty-eight hundredths**. **b. 2.6**. "Two" is the whole number, "and" is the point, 6 goes in the tenths place.

F 0.3 = 0.30, and 30 hundredths > 27 hundredths, so **0.3 > 0.27**.
On the grids: 3 full columns for 0.3; 2 columns + 7 squares for 0.27.



0.3 0.27

G Each jump is 0.1 (one tenth). P is 4 jumps from 0, so P = **0.4**.

H **0.236**. 2 full columns = 2 tenths; 3 more squares = 3 hundredths; 6 slices = 6 thousandths. So 0.236 is a little more than 23 hundredths.

On Your Own: 1–9

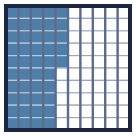
1 **C**, one thousandth. 1,000 equal parts are thousandths. (10 parts are tenths; 100 parts are hundredths.)

2 **0.608**. 6 full columns = 6 tenths. No extra squares, so 0 hundredths (a placeholder zero). 8 slices = 8 thousandths.

3 **B**, 0.3. 3 of 10 equal parts are shaded. D (0.7) is the **unshaded** part; A (0.03) would be 3 hundredths.

4 **0.07**. From 0 to 0.1 is 1 tenth = 10 hundredths, so each jump is 0.01. A is 7 jumps from 0.

5 Shade **4 full columns** and **5 more squares** (45 squares in all). 0.45 = 4 tenths + 5 hundredths.



6 **C**, the 3. In 5.438: 5 ones, 4 tenths, **3 hundredths**, 8 thousandths.

7 **0.09** (9 hundredths). In 0.392 the 9 is the second digit after the point, the hundredths place. *Not* 9 or 0.9.

8 **a. 0.604**. **b. 0.6 + 0.004**. Chart: 0 . 6 0 4. The 0 holds the hundredths place. Without it, 0.64 would mean 6 tenths and 4 hundredths, a different number.

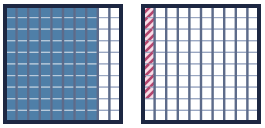
9 **two hundred nine thousandths**. Read 209 as a whole number, then say the place of the last digit (thousandths). There is no "and," because there is no whole number.

On Your Own: 10–17

10 **3.15**. "Three" is the whole number, "and" is the decimal point, and fifteen hundredths needs 2 places: .15.

11 **B and C**. 0.60 and 0.600 add zeros at the **end**, which does not change the value. A (0.06) is 6 hundredths, which is smaller. D (6.0) is 6 wholes.

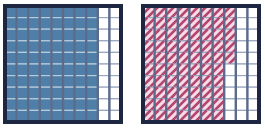
12 **No, Mia is not correct.** 0.8 is 8 tenths (80 hundredths). 0.08 is 8 hundredths. The zero in 0.08 holds the tenths place, so the 8 is worth only one tenth as much. A zero added at the **end**, after the last decimal digit, does not change the value: 0.8 = 0.80. *Full credit: says "no" and gives a place-value or model reason.*



0.8 0.08

13 **a. > b. = c. < d. <**
a. 0.70 vs 0.69: 70 hundredths > 69 hundredths. b. 0.25 = 0.250 (a zero at the end). c. 0.04 is 4 hundredths; 0.4 is 4 tenths. d. 0.512 vs 0.520: the tenths are equal (5); hundredths 1 < 2. Not "512 > 52"!

14 **Jordan ran farther.** 0.8 = 0.80 = 80 hundredths, and 80 hundredths > 75 hundredths. Or compare tenths: 8 tenths > 7 tenths. Ava compared 75 and 8 as whole numbers, but the digits are in different places. *Full credit: correct runner and a place-value reason.*



0.8 0.75

15 **0.6**. 10 − 4 = 6 pieces are left. 6 of 10 equal pieces is 6 tenths = 0.6. (0.4 is the part Sam **ate**.)

16 **10 hundredths, 100 thousandths**. One column of the grid holds 10 hundredths, and each hundredth holds 10 thousandths: 10 × 10 = 100.

17 **0.702**. Thousandths = 2, hundredths = 0, tenths = 2 + 5 = 7. It is less than 1, so the ones digit is 0.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
0	.	7	0	2

MY FIX-IT NOTES

For each question I missed: the question number, what I did wrong, and what I will remember next time.

Scoring On Your Own: 17 questions; 13 parts a–d count as one question (all four correct). 8a and 8b count as one question (both correct). Questions 12 and 14 need a reason to count as correct.

Parent Mastery Checklist

Use the On Your Own results to decide what is mastered and what needs more practice.

Skill (same as the Math Lab app)	Questions	Signs of mastery	Mastered	Practice more
Equal parts: tenths, hundredths, thousandths	1, 16	Names the part for 10, 100, and 1,000 equal parts; knows 10 of one place make 1 of the next.	☐	☐
Decimal models: bars, grids, number lines	2, 3, 4, 5, 15	Counts columns as tenths, squares as hundredths, and slices as thousandths; finds the jump size on a number line.	☐	☐
Place value of decimal digits	6, 7, 8, 17	Names the place and value of any digit (the 9 in 0.392 is 0.09); uses placeholder zeros.	☐	☐
Reading and writing decimals	9, 10	Reads with place names (“two hundred nine thousandths”); uses “and” only for the decimal point.	☐	☐
Equivalent decimals and zero placeholders	11, 12	Knows 0.6 = 0.60 = 0.600 but 0.06 ≠ 0.6; explains why with place value.	☐	☐
Comparing with place value	13, 14	Writes equal numbers of places or compares tenths first; is not fooled by “more digits.”	☐	☐

READY FOR LESSON 2?

A good sign of independent mastery is **15 or more of the 17** On Your Own questions correct, including reasons for 12 and 14. That is about the same as the app's suggestion of 9 of 10 on its D01 Test. *This is a family benchmark, not an official Georgia score.*

Score: _____ / 17 Date: _____

COMMON MISTAKES TO WATCH FOR

- Thinking more digits means bigger (0.35 > 0.4).
- Dropping a placeholder zero (0.07 written as 0.7).
- Counting the **unshaded** part of a model.
- Mixing up hundredths and thousandths places.
- Reading “point four three” without the place name; ask for “forty-three hundredths.”

TALK ABOUT IT (2 MINUTES)

- “Show me 0.3 and 0.03 on a grid. Which is bigger? How do you know?”
- “Why is 0.5 the same as 0.50, but 0.05 is different?”
- “What is the value of the 4 in 1.246?” (0.04)

NEXT STEPS

- **Mastered:** do the D01 **Test** in Gabriella's Math Lab, then move on to Lesson 2: Decimal Place Value.
- **Practice more:** reread the matching See It page, then do a D01 **Practice** session in the app.

Georgia standards: 5.NR.4.1 and 5.NR.4.2, connecting to 5.NR.1.1. Comparing is introduced here and taught fully in Lesson 4.